Methodology-Faithful Reconstruction and D0 Sensitivity Analysis of Q-Learning-Based Routing in a 3D Wireless Sensor Network

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***Abstract*—**This paper presents a methodology-faithful reconstruction and experimental validation of a Q-learning-based energy-efficient routing scheme for three-dimensional Wireless Sensor Networks (WSNs). Because the complete original network topology, coordinate table, and several simulator parameters were unavailable, this work reconstructs the reward function, radio energy model, and propagation model of the reference method and validates them through packet-level, link-level, and full-network experiments on a reconstructed 45-node, three-dimensional deployment. The reconstructed system was validated through single-packet reward/Q-value/energy audits, a controlled infeasible-route rejection test, and a full lifetime experiment that completed 9189 rounds and delivered 4704.77 Kbits before terminating due to exhausted feasible routes, with all 44 sensor nodes remaining alive. These results are compared against the published proposed Q-learning result (6260 rounds, 3205.12 Kbits) and the published Dijkstra baseline (2737 rounds, 1401.34 Kbits). An additional sensitivity study varying the distance threshold D0 from 2 m to 6 m shows that this reconstruction assumption materially, and non-monotonically, affects network lifetime. The results are presented as a validated reconstruction of the core methodology rather than an exact numerical replication, since several topology and simulation details had to be assumed.

Keywords*—Wireless Sensor Networks, Q-learning, energy-efficient routing, three-dimensional network, reproducibility, reinforcement learning*

I. Introduction

Wireless Sensor Networks (WSNs) consist of spatially distributed, battery-powered sensor nodes that must relay data toward a sink while operating under severe energy constraints. Because node replacement or recharging is often impractical, energy-aware routing is central to extending network lifetime. Reinforcement learning, and Q-learning in particular, has been proposed as a mechanism for adaptive routing that jointly accounts for residual energy, distance, and hop count when selecting a next-hop sensor.

This paper reports a methodology-faithful reconstruction of a Q-learning-based energy-efficient routing scheme for three-dimensional WSNs, together with an experimental validation of that reconstruction on a 45-node deployment. The complete original node-coordinate table, communication topology, and certain simulation parameters were not available, so this work explicitly separates parameters taken directly from the reference description from parameters that were assumed or reconstructed for the purpose of running a complete, testable implementation. The objective is not to claim an exact numerical replication of the original results, but to demonstrate that the core methodology — the reward formulation, the radio energy model, the propagation model, and the Q-learning update — can be reconstructed, implemented, and experimentally validated in a manner consistent with the reference description.

The remainder of this paper is organized as follows. Section II summarizes related work. Section III formulates the routing and energy problem. Section IV describes the reconstructed methodology. Section V describes the experimental setup. Section VI presents packet-level, link-level, and full-network results and compares them against published values. Section VII presents a D0 sensitivity extension. Section VIII discusses limitations, and Section IX concludes the paper.

II. Related Work

Energy-aware routing in WSNs has been studied extensively, ranging from clustering-based protocols such as LEACH [2], to shortest-path approaches such as Dijkstra's algorithm [4], to learning-based approaches that adapt routing decisions to residual energy and network state. Q-learning [3], a model-free reinforcement learning method, is well suited to this setting because it allows a routing policy to be refined online from local reward signals without requiring a full model of network dynamics. The reference method this work reconstructs formulates next-hop selection as a Q-learning problem in which the reward combines residual energy, distance, and remaining hop count, and compares the resulting routing performance against a Dijkstra shortest-path baseline. The exact original publication is cited in Reference [1]; a complete bibliographic citation should be inserted prior to submission.

III. Problem Formulation

Consider a WSN deployed in a three-dimensional volume consisting of N sensor nodes and a single sink. Each sensor node has finite initial energy and must relay data packets toward the sink, either directly or through intermediate sensors. At each routing decision, a candidate next-hop node is evaluated according to its residual energy, its distance to the current node, and the number of hops remaining to the sink. The routing objective is to select, for each transmission, a feasible path that balances energy expenditure across the network so as to maximize the number of successful transmissions before any node's energy is exhausted, while never selecting a route that would drive a node's energy below zero.

IV. Methodology

A. Q-Learning Formulation

The reconstructed implementation evaluates candidate routes using a reward and path-quality formulation and updates a Q-value for each candidate accordingly. The Q-value update rule implemented is:

$$Q\_new = (1 - \alpha) \cdot Q\_old + \alpha \cdot [Reward + PathQuality] \quad (1)$$

where α is the learning rate, Reward is computed as described in Section IV-B, and Path Quality is a route-level score based on the residual energy and remaining hop count of candidate next-hop sensor nodes.

B. Reward Function

For an intermediate next-hop sensor, the implemented reward is the residual energy of the candidate node divided by the propagation-weighted distance and the remaining hop count:

$$Reward = E\_residual / (distance^{\beta} \times hop\_count)$$

The sink node does not contribute a battery-energy reward term, since it is not energy-constrained in the same sense as sensor nodes.

C. Radio Energy Model

Transmission and reception energy follow a first-order radio energy model. Transmission energy for a k-bit packet over distance d is:

$$E\_TX(k,d) = E\_elec \cdot k + E\_amp \cdot k \cdot d^{\beta}$$

and reception energy is:

$$E\_RX(k) = E\_elec \cdot k$$

The implementation uses $E\_elec = 5 \times 10^{-8}$ J/bit, $E\_amp = 1 \times 10^{-10}$ J/bit/m², and a fixed packet size of 512 bits, consistent with commonly used first-order radio energy models in the WSN literature [2].

D. Propagation Model

The path-loss exponent β is set according to a distance threshold D0: $\beta = 2$ for distances at or below D0, and $\beta = 4$ for distances beyond D0, reflecting the transition between free-space and multipath propagation regimes. In this reconstruction, D0 = 4.0 m is used as an implementation assumption; it is not claimed to be a numerical parameter taken directly from the original reference.

V. Experimental Setup

A. Network Configuration

The reconstructed experiment uses a 45-node deployment consisting of 44 sensor nodes and a single sink, deployed within a 16 × 13 × 2.5 m volume. Sensor coordinates were reconstructed using a fixed random seed (42) because the complete original coordinate table was not available; a coordinate audit confirmed 45 coordinate entries within the deployment volume, with the sink placed at (8.0, 6.5, 0). Table I summarizes the frozen reconstruction parameters together with their provenance, distinguishing values taken from the reference setup, values derived from the radio energy model, and values that were assumed for this reconstruction.

Parameter	Value	Provenance
Total nodes	45	Paper/reference setup
Sensor nodes	44	Paper/reference setup
Sink	1	Paper/reference setup
Deployment volume	16 × 13 × 2.5 m	Paper/reference setup
Initial energy	0.5 J/node	Paper/reference setup
Packet size	512 bits	Paper/reference setup
E_elec	5×10^{-8} J/bit	Energy model
E_amp	1×10^{-10} J/bit/m²	Energy model
Coordinate seed	42	Our reconstruction
Communication range	7.0 m	Our assumption
D0	4.0 m	Our assumption
Source scheduling	Round-robin (alive/routable)	Our implementation
Maximum hops	6	Routing-table limit
Maximum stored paths	20 / source	Routing-table limit

TABLE I. FROZEN RECONSTRUCTION PARAMETERS AND PROVENANCE

B. Routing-Table Construction

For each sensor, the implementation precomputes and stores loop-free simple paths to the sink, bounded by a maximum of six hops and twenty stored candidate paths per source. At each Q-learning round, the routing procedure retrieves the precomputed candidate paths for the active source and evaluates each candidate using link distance, hop count, reward, path quality, and energy feasibility.

C. Energy-Feasibility and Safety Corrections

Two corrections were introduced to ensure physically valid energy accounting. First, direct transmit/receive energy-decrement operations were replaced with a bounded deduction procedure that prevents any node from being assigned a negative energy value. Second, before a route is selected, the required transmit and receive energy is aggregated per sensor across the complete candidate path, and a candidate is rejected if any node along the path cannot afford its total energy requirement; this prevents a route from passing a partial feasibility check and later failing mid-transmission. A controlled feasibility test, in which a node was given only 0.000060 J against a route requiring approximately 0.00008498 J, confirmed that infeasible routes are correctly rejected.

VI. Results and Discussion

A. Single-Packet Validation

A clean single-packet run was used to validate the reward, Q-value, and energy computations end to end. Table II reports the selected path and the resulting reward, path quality, updated Q-value, and energy consumption. No node reached a negative-energy state, and the minimum residual energy across the path remained well above zero.

Metric	Result
Selected path	S1→S3→S6→S7→S8→S17→Sink
Reward	0.3611352756
Path quality	1.1416666667
New Q	0.7514009711
Energy used	0.000422052451 J
Negative-energy nodes	None
Minimum residual energy	0.499889424352 J

TABLE II. CLEAN ONE-PACKET VALIDATION

B. Link-by-Link Energy Audit

The selected packet route was independently audited link by link, comparing manually computed transmit and receive energies against the values produced by the implementation. able III shows that the manually calculated and implementation-computed TX/RX energy values agreed to the displayed numerical precision for all audited links.

Link	d (m)	β	TX (J)	RX (J)	Total (J)
S1→S3	4.1397	4	0.000040636747	0.000025600000	0.000066236747
S3→S6	5.8356	4	0.000084975648	0.000025600000	0.000110575648
S6→S7	5.0928	4	0.000060043000	0.000025600000	0.000085643000
S7→S8	4.9638	4	0.000056682702	0.000025600000	0.000082282702
S8→S17	1.1785	2	0.000025671106	0.000025600000	0.000051271106
S17→Sink	2.9423	2	0.000026043248	0	0.000026043248

TABLE III. LINK-LEVEL ENERGY AUDIT

C. Final 45-Node Q-Learning Experiment

The full reconstructed experiment ran until no further feasible packet could be selected. Table IV summarizes the final results: the experiment completed 9189 successful packet rounds, transmitting a total of 4,704,768 bits (4,704.77 Kbits) while consuming 4.080833 J of total network energy. All 44 sensor nodes remained alive at termination; the run ended because the complete-path feasibility check could not identify any further usable route, not because of node death.

Metric	Result
Rounds completed	9189
Successful packets	9189
Total transmitted bits	4,704,768
Total transmitted Kbits	4,704.77
Total energy consumed	4.080833 J
Final residual energy	17.919167 J
Alive nodes	44
Dead nodes	0
Minimum residual energy	0.000020122862 J
Execution time	9.71 s

TABLE IV. FINAL RECONSTRUCTED 45-NODE EXPERIMENT

The lowest-energy nodes at termination were S3 (0.000020122862 J), S2 (0.000020960907 J), and S1 (0.000050827213 J). Figs. 1 and 2 illustrate the corresponding residual-energy variation and number of alive sensor nodes throughout the reconstructed Q-learning experiment.

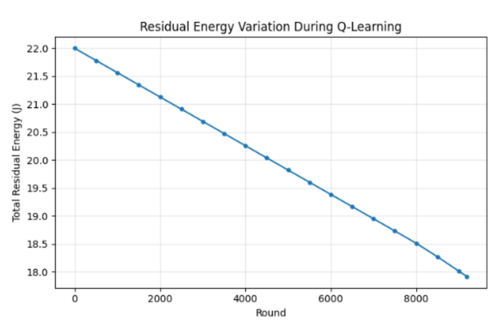

Fig. 1. Residual Energy Variation During Q-Learning Experiment.

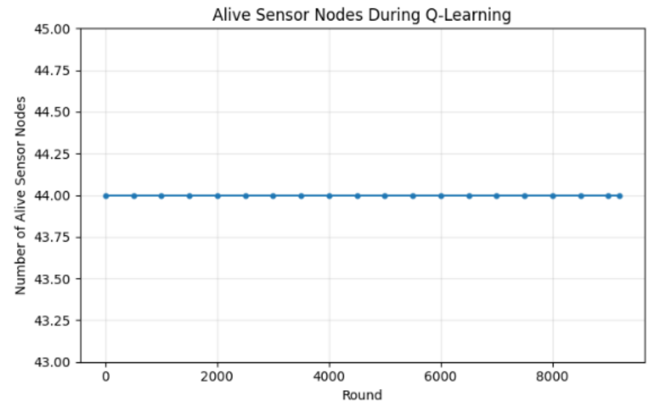

Fig. 2. Number of Alive Sensor Nodes During Q-Learning Experiment.

D. Comparison with Published Results

Table V compares the reconstructed result against the published proposed Q-learning result and the published Dijkstra baseline for the same 45-node, three-dimensional scenario. The reconstructed implementation achieves a longer lifetime and delivers more data than the published Q-learning result. This is not interpreted as an improvement over the original method, since the exact original topology, coordinate table, and several simulation parameters were unavailable and had to be reconstructed or assumed; rather, it indicates that the reconstructed methodology behaves consistently with, but not numerically identically to, the reference method under the stated assumptions. Table V summarizes the numerical comparison, while Figs. 3 and 4 provide visual comparisons of the completed transmission rounds and total conveyed data, respectively.

Result	Lifetime (rounds)	Data (Kbits)
Published proposed Q-learning	6260	3205.12
Published Dijkstra	2737	1401.34
Our reconstructed Q-learning	9189	4704.77

TABLE V. PUBLISHED VALUES VERSUS RECONSTRUCTED RESULT

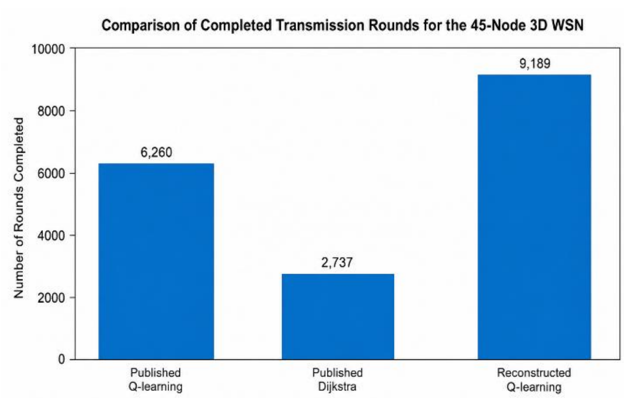

Fig. 3. Comparison of Completed Transmission Rounds for the 45-Node 3D WSN.

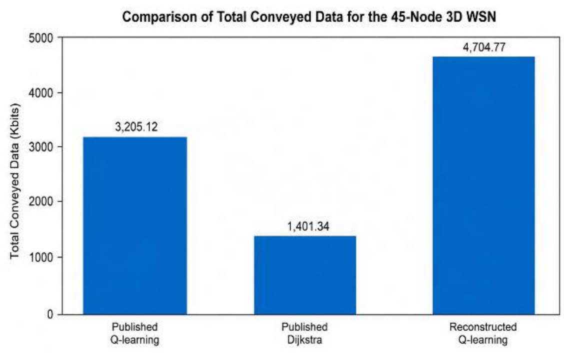

Fig. 4. Comparison of Total Conveyed Data for the 45-Node 3D WSN.

VII. D0 Sensitivity Extension

Beyond the frozen baseline, an additional experiment varied the propagation distance threshold D0 from 2 m to 6 m, with each run starting from fresh 0.5 J per-sensor energy and the baseline state restored afterward. Table VI reports network lifetime, delivered data, energy consumed, and final residual energy for each value of D0. The results show that D0 materially affects routing performance, but the relationship is not strictly monotonic: D0 = 3 m yields slightly fewer rounds than D0 = 2 m, while D0 = 5 m and D0 = 6 m yield progressively longer lifetimes. This extension characterizes the sensitivity of the reconstruction to one of its key assumed parameters, rather than reporting a result from the original reference.

D0 (m)	Rounds	Data (Kbits)	Energy consumed (J)	Residual energy (J)
2	9088	4653.056	4.164968	17.835032
3	9059	4638.208	4.066465	17.933535
4	9189	4704.768	4.080833	17.919167
5	9512	4870.144	4.076632	17.923368
6	10218	5231.616	3.835497	18.164503

TABLE VI. D0 SENSITIVITY RESULTS

A communication-range sensitivity experiment (5–9 m) was also attempted, but the initial results were invalidated by a routing-table inconsistency in which changing the network graph did not propagate to the precomputed path structure used during Q-learning rounds. Following correction of the route-generation procedure, communication-range results are withheld from this paper until the corrected implementation is verified to reproduce the validated 9189-round baseline at the 7 m configuration.

VIII. Limitations

This work reconstructs and validates the core Q-learning routing methodology but does not achieve an exact numerical replication of the original published results. The following elements were not available from the original reference and were therefore assumed or reconstructed: the exact original node-coordinate dataset; the guaranteed original communication topology; the numerical value of D0 (assumed as 4 m); the numerical communication range (assumed as 7 m); the exact original source-scheduling procedure; and the complete original routing-table generation procedure. Consequently, the published lifetime of 6260 rounds should not be treated as a target that this reconstruction is expected to reproduce exactly. Comparison figures use published final values only where round-by-round published data were unavailable, and no unsupported intermediate curves were invented for the published baselines.

IX. Conclusion

This paper presented a methodology-faithful reconstruction and experimental validation of a Q-learning-based energy-efficient routing scheme for a 45-node, three-dimensional Wireless Sensor Network. The reconstructed reward function, radio energy model, propagation model, and Q-learning update were validated through single-packet, link-level, and full-network experiments, including a controlled test confirming correct rejection of energy-infeasible routes. The full reconstructed experiment completed 9189 rounds and delivered 4704.77 Kbits with all 44 sensor nodes remaining alive at termination, and a D0 sensitivity study showed that this reconstruction assumption materially and non-monotonically affects network lifetime. The principal limitation of this study is the unavailability of the complete original topology and certain simulation parameters, which prevents exact numerical reproduction of the published results. The appropriate characterization of this work is a validated reconstruction of the core methodology, not an exact replication, and future work should focus on reproducing the original topology more closely, validating additional network configurations, and extending the sensitivity analysis to other simulation parameters.

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